

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining

Date: **May 9, 2017**

Time: 6:30-8:30pm

Student ID: _____

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Calculator Brand: _____ Calculator Model: _____

Answer ALL 7 Questions.

Answers should be written on the answer book.

Question No.	Mark
Question 1 (10%)	
Question 2 (15%)	
Question 3 (15%)	
Question 4 (15%)	
Question 5 (15%)	
Question 6 (15%)	
Question 7 (15%)	
Total (100%)	

Question 1. (attribute types, 10%). Give an example for each of the following attribute types: nominal, ordinal, continuous ordinal.

Question 2. (Apriori Algorithm, 15%). Table 1 shows a list of transactions in a restaurant in Italy. Each transaction contains the list of the ingredients of a "pizza" ordered by a customer in that restaurant: tomato (T), mozzarella (M1), basil (B), olives (O), Ham (H) and mushrooms (M2). For example transaction 1 indicates that a customer ordered one pizza with mozzarella, olives and mushrooms. Using the Apriori algorithm compute the frequent itemsets in the transactions in Table 1 with support threshold $s=2$. For each step of the algorithm, specify the counters that are kept in main memory as well as the frequent itemsets found in that step.

Transaction num.	Ingredients
1	M1,O,M2
2	T,M1,B
3	T,M1,B
4	T,M1
5	T,O
6	T,M1,H,O
7	T,H,M1

Table 1. List of transactions in a restaurant in Italy. For each transaction the list of ingredients of the corresponding "pizza" are specified.

Question 3. (SON Algorithm, 15%). Table 2 shows a list of transactions for the following set of items: {a,b,c,d,e,f,g,h,l}. Suppose that such a list is partitioned into three different machines M1,M2,M3 as follows. M1 receives the first four itemsets i.e. {a},{a,b},{c,d},{d,e}, M2 receives the next four itemsets i.e. {b,c},{e,f},{f,g},{a,c,d}, while M3 receives the remaining three itemsets, i.e. {b},{h},{c,d,l}. Using the SON algorithm, compute the frequent itemsets in such a list of transactions with support threshold $s=3$. In particular, for each machine specify the support threshold used in that machine as well as the frequent itemsets found in that machine (with the corresponding threshold).

Specify the list of frequent itemsets in the whole list, which have been found at the end of the SON algorithm. In order to compute the list of frequent itemsets in each machine, you can use any algorithm. You are requested to specify only the list of frequent itemsets in each machine, intermediary steps of the algorithm used for this task can be omitted.

Transaction num.	Items
1	a
2	a,b
3	c,d
4	d,e
5	b,c
6	e,f
7	f,g
8	a,c,d
9	b
10	h
11	c,d,l

Table 2. List of transactions and the corresponding list of items.

Question 4. (K-Means, 15%) Suppose we are given the following points in the Euclidean space: $P1=(0,0)$, $P2=(0,1)$, $P3=(1,1)$, $P4=(3,0)$, $P5=(4,0)$, $P6=(4,1)$. Suppose that $P1$ and $P2$ are selected as the initial set of centroids for the K-means algorithm, with $K=2$. Run the K-means algorithm on those input points, while showing all the steps.

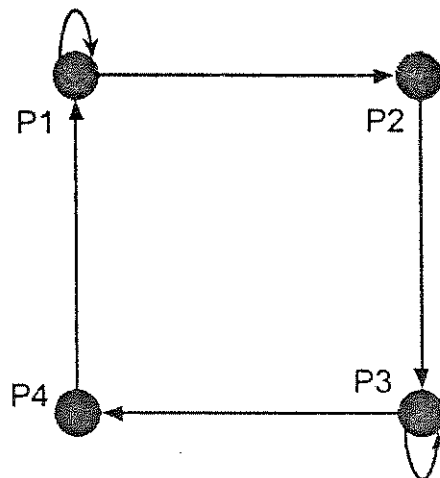


Figure 1. A simple Web graph with four Web pages: $P1, P2, P3, P4$.

Question 5. (PageRank, 15%) Compute the PageRank vector of the graph depicted in Figure 1. In particular, write down 1) the matrix M representing the Markov Chain associated with such a graph 2) the corresponding system of linear equations and solve it. Show the first two steps of the PageRank algorithm, starting with the vector $(1/4, 1/4, 1/4, 1/4)$. Argue that the PageRank algorithm with input the graph in Figure 1 converges to the same vector, regardless of the starting vector.

Question 6. (Decision trees, 15%) Given the data in Table 3, construct a decision tree using the following rules: (i) at each step compute the gini index for every possible split considering all attributes (except 'Animal') and select the split with the best gini value; (ii) stopping rule: when the gini value of a node is zero or no further split is possible; (iii) the class of a leaf node is determined by the majority rule (breaking ties arbitrarily). After having built the decision tree, determine whether the tree can be pruned, so as to improve the generalization error. Recall that the training errors are given by the number of records that are misclassified by the decision tree, while the generalization error is equal to the number of training errors plus 0.5 times the number of leaves. Each subtree should be pruned (i.e. replaced by a leaf node) until no further improvement is possible. Each leaf node is then labeled by the majority rule (breaking ties arbitrarily). Show all the steps. Predict the class value of the record: ('seal', GB='Y', LW='Y') using the decision tree which minimizes the generalization error

Animal	Give Birth	Live in Water	Class
whale	Y	Y	M
dolphin	Y	Y	M
orca	Y	Y	M
human	Y	N	M
boa snake	Y	N	NM
Surinam toad	Y	N	NM
owl	N	N	NM
platypus	N	N	M
salmon	N	Y	NM
eel	N	Y	NM

Table 3. The table specifies for each animal whether it gives birth, lives in water and is mammal or non-mammal.

Question 7. (Naive Bayes, 15%) Given the data in Table 4, use the Naive Bayes classifier to predict the class of the record: ('salmon', GB='N', LW='Y'). Show all steps.

Animal	Give Birth	Live in Water	Class
whale	Y	Y	M
echidna	N	N	M
human	Y	N	M
Boa snake	Y	N	NM
owl	N	N	NM
salmon	N	Y	NM
Viper snake	Y	N	NM

Table 4. The table specifies for each animal whether it gives birth, lives in water and is mammal or non-mammal.

END OF PAPER